Chapter 1

Discussion

This chapter summarizes the contributions of this thesis, articulates some of the questions raised by this work, and relates the kernel-based model-building procedure of section 1.4 to the larger project of automating statistics and machine learning.

1.1 Summary of contributions

The main contribution of this thesis was to develop a way to automate the construction of structured, interpretable nonparametric regression models using Gaussian processes. This was done in several parts: First, section 1.4 presented a systematic overview of kernel construction techniques, and examined the resulting GP priors. Next, section 1.4 showed the viability of a search over an open-ended space of kernels, and showed that the corresponding GP models could be automatically decomposed into diverse parts allowing visualization of the structure found in the data. Section 1.4 showed that sometimes parts of kernels can be described modularly, allowing automatically written text to be included in detailed reports describing GP models. An example report is included in appendix **??**. Together, these chapters demonstrate a proof-of-concept of what could be called an "automatic statistician" capable of performing some of the model construction and analysis currently performed by experts.

The second half of this thesis examined several extensions of Gaussian processes, all of which enabled the automatic determination of model choices that were previously set by trial and error or cross-validation. Section 1.4 characterized and visualized deep Gaussian processes, related them to existing deep neural networks, and derived novel deep kernels. Section 1.4 investigated additive GPs, and showed that they have the same covariance as a GP using dropout. Section 1.4 extended the GP latent variable model

into a Bayesian clustering model which automatically infers the nonparametric shape of each cluster, as well as the number of clusters.

1.2 Structured versus unstructured models

One question left unanswered by this thesis is: when to prefer the structured, kernel-based models described in section 1.4 to the relatively unstructured deep GP models described in section 1.4? This section considers some advantages and disadvantages of the two approaches.

- **Difficulty of optimization.** The discrete nature of the space of composite kernel structures can be seen as both a blessing and a curse. Certainly, a mixed discrete-and-continuous search space requires more complex optimization procedures than the continuous-only optimization possible in deep GPs.

  However, the discrete nature of the space of composite kernels offers the possibility of learning heuristics to suggest which types of structure to add, based on features of the dataset, or previous model fits. For example, finding periodic structure or growing variance in the residuals suggests adding periodic or linear components to the kernel, respectively. It is not clear whether such heuristics could easily be constructed for optimizing the variational parameters of a deep GP.

- **Long-range extrapolation.** Another open question is whether the inductive bias of deep GPs can be made to allow the sorts of long-range extrapolation shown in section 1.4. As an example, consider the problem of extrapolating a periodic function. A deep GP could learn a latent representation similar to that of the periodic kernel, projecting into a basis equivalent to $[\sin(x), \cos(x)]$ in the first hidden layer. However, to extrapolate a periodic function, the sin and cos functions would have to repeat beyond the input range of the training data, which would not happen if each layer assumed only local smoothness.

  One obvious possibility is to marry the two approaches, building deep GPs with structured kernels. However, we may lose some of the advantages of interpretability by this approach, and inference would become more difficult.

  Another point to consider is that, in high dimensions, the distinction between interpolation and extrapolation becomes less meaningful. If the training and test data both live on a low-dimensional manifold, then learning a suitable representation of that manifold may be sufficient for obtaining high predictive accuracy.

- **Ease of interpretation.** Historically, the statistics community has put more emphasis on the interpretability and meaning of models than the machine learning community, which has focused more on predictive performance. To begin to automate the practice of statistics, developing model-description procedures for powerful open-ended model classes seems to be a necessary step.

  At first glance, automatic model description may seem to require a decomposition of the model being described into discrete components, as in the additive decomposition demonstrated in section 1.4.

  On the other hand, most probabilistic models allow a form of summarization of high-dimensional structure through sampling from the posterior. For the specific case of deep GPs, Damianou and Lawrence (2013) showed that these models allow summarization through examining the dimension of each latent layer, visualizing latent coordinates, and examining how the predictive distribution changes as one moves in the latent space. Perhaps more sophisticated procedures could also allow intelligible text-based descriptions of such models.

The warped mixture model of section 1.4 represents a compromise between these two approaches, combining a discrete clustering model with an unstructured warping function. However, the explicit clustering model may be unnecessary: the results of Damianou and Lawrence (2013) suggest that clustering can be automatically (but implicitly) accomplished by a sufficiently deep, unstructured GP.

1.3 Approaches to automating model construction

This thesis is a small part of a larger push to automate the practice of model building and inference. Broadly speaking, this problem is being attacked from two directions.

From the top down, the probabilistic programming community is developing automatic inference engines for extremely broad classes of models (Goodman et al., 2008; Koller et al., 1997; Mansinghka et al., 2014; Milch et al., 2007; Stan Development Team, 2014; Wood et al., 2014) such as the class of all computable distributions (Li and Vitányi, 1997; Solomonoff, 1964). As discussed in **??**, model construction procedures can usually be seen as a search through an open-ended model class. Exhaustive search strategies have been constructed for the space of computable distributions (Hutter, 2002; Levin, 1973; Schmidhuber, 2002), but they remain impractically slow.

An alternative, bottom-up, approach is to design procedures which extend and combine existing model classes for which relatively efficient inference algorithms are already known. The language of models proposed in section 1.4 is an example of this bottom-up approach. Another example is Grosse (2014), who built an open-ended language of matrix decomposition models and a corresponding compositional language of relatively efficient approximate inference algorithms. Similarly, Steinruecken (2014) showed how to compose inference algorithms for sequence models. These approaches have the advantage that inference is usually feasible for any model in the language. However, extending these languages may require developing new inference algorithms.

If sufficiently powerful building blocks are composed, the line between the top-down and bottom-up approaches becomes blurred. For example, deep generative models (Adams et al., 2010; Bengio et al., 2013; Damianou and Lawrence, 2013; Rippel and Adams, 2013; Salakhutdinov and Hinton, 2009) could be considered an example of the bottom-up approach, since they compose individual model "layers" to produce more powerful models. However, large neural nets can capture enough different types of structure that they could also be seen as an example of the universalist, top-down approach.

1.4 Conclusion

It seems clear that one way or another, large parts of the existing practice of model-building will eventually be automated. However, it remains to be seen which of the above model-building approaches will be most useful. I hope that this thesis will contribute to our understanding of the strengths and weaknesses of these different approaches, and towards the use of more powerful model classes by practitioners in other fields.

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